

1. AIM

To design and verify the schematic and layout of a Dynamic CMOS inverter using 180 nm technology, and compare its pre-layout and post-layout simulation results

2. TOOL

Cadence Virtuoso

3. THEORY

A Dynamic CMOS inverter operates in two distinct phases: precharge and evaluation, controlled by a clock signal.

During the precharge phase, when the clock is LOW, the precharge PMOS transistor is ON and charges the output node to V_{dd} . The pull-down network (PDN) is disabled during this phase, so the output is precharged to logic HIGH.

During the evaluation phase, when the clock becomes HIGH, the precharge transistor is turned OFF and the PDN is enabled. For a LOW input, the NMOS transistor remains OFF and the output retains its precharged HIGH state. For a HIGH input, the NMOS transistor turns ON and provides a discharge path from the output node to ground, causing the output to transition to LOW.

The propagation delays are defined as t_{pLH} for a LOW-to-HIGH transition and t_{pHL} for a HIGH-to-LOW transition. Since the output is already precharged to HIGH before the evaluation phase, no LOW-to-HIGH transition occurs during evaluation. Therefore, $t_{pLH} = 0$ for the evaluation operation.

The schematic of the dynamic CMOS inverter is designed and simulated using 180 nm CMOS technology in Cadence Virtuoso. The corresponding physical layout is designed and verified using design-rule checking (DRC) and layout-versus-schematic (LVS) verification.

Pre-layout and post-layout simulations are then performed and compared to analyze the effect of layout parasitics on propagation delay and overall circuit performance.

4. TABLES

Table 4.1: CMOS Inverter Design Parameters

Parameter	Value	Unit
Technology	180	nm
V_{dd}	1.8	V
PMOS Width	400	nm
NMOS Width	400	nm

Table 4.2: Result comparision (pre layout vs post layout)

Parameter	Pre layout	Post layout	Unit	Error(wiht respect to pre layout)(%)
V_{th}	715.711	715.7106	mV	0.0001%
V_{OH}	1.8	1.8	V	0.00%
V_{OL}	0	0	V	0.00%
V_{IH}	836.654	836.9212	mV	0.0319%
V_{IL}	555.495	555.4912	mV	0.0007%
NM_H	0.9633	0.9631	V	0.0208%
NM_L	555.495	555.4912	V	0.0007%
t_{HL}	14.4	22.7	ps	57.6389%
t_{LH}	0	0	ps	0.00%
t_p	14.4	22.7	ps	57.6389%
t_r	46.3	80	ps	72.7862%
t_f	23.9	40.6	ps	69.8745%

Table 4.3: Result comparision (static vs dynamic for pre layout)

Parameter	Static	Dynamic	Unit
V_{th}	713.958	715.711	mV
V_{OH}	1.8	1.8	V
V_{OL}	0	0	V
V_{IH}	827.4818	836.654	mV
V_{IL}	541.4587	555.495	mV
NM_H	0.9725	0.9633	V
NM_L	541.4587	555.495	V
t_{HL}	15.02	14.4	ps
t_{LH}	33.7	0	ps
t_p	15.02	14.4	ps
t_r	28.89	46.3	ps
t_f	20.42	23.9	ps

Table 4.4: Result comparison (static vs dynamic for post layout)

Parameter	Static	Dynamic	Unit
V_{th}	712.969	715.7106	mV
V_{OH}	1.8	1.8	V
V_{OL}	0	0	V
V_{IH}	827.543	836.9212	mV
V_{IL}	541.4642	555.4912	mV
NM_H	0.9725	0.9631	V
NM_L	0.5415	0.5555	V
t_{HL}	22.05	22.7	ps
t_{LH}	48.06	0	ps
t_p	22.05	22.7	ps
t_r	78.75	80	ps
t_f	28.33	40.6	ps

5. CALCULATION

$$N_{MH} = V_{OH} - V_{IH} = 1.8 - 0.836654 = 0.963346V \approx 0.9633V$$

$$N_{ML} = V_{IL} - V_{OL} = 0.555495 - 0 = 0.555495V$$

$$N_{MH(post)} = V_{OH} - V_{IH} = 1.8 - 0.8369212 = 0.9630788V \approx 0.9631V$$

$$N_{ML(post)} = V_{IL} - V_{OL} = 0.5554912 - 0 = 0.5554912V$$

$$t_{HL(pre)} = (37.0219 - 37.0075) * 1000 = 14.4ps$$

$$t_{HL(post)} = (37.0302 - 37.0075) * 1000 = 22.7ps$$

$$t_{LH(pre)} = 0ps$$

$$t_{LH(post)} = 0ps$$

$$t_{p(pre)} = 37.0219ps$$

$$t_{p(post)} = 37.0302ps$$

$$t_{r(pre)} = (20.0648 - 20.0185) * 1000 = 46.3ps$$

$$t_{r(post)} = (20.1034 - 20.0234) * 1000 = 80.0ps$$

$$t_{f(pre)} = (37.0376 - 37.0137) * 1000 = 23.9ps$$

$$t_{f(post)} = (37.056 - 37.0154) * 1000 = 40.6ps$$

$$Error_{V_{th}} = (0.0004/15.711) * 100 = 0.0001$$

$$Error_{V_{IH}} = (0.2672/836.654) * 100 = 0.0319$$

$$Error_{V_{IL}} = (0.0038/555.495) * 100 = 0.0007$$

$$Error_{N_{MH}} = (0.0002/0.9633) * 100 = 0.0208$$

$$Error_{N_{ML}} = (0.0000038/0.555495) * 100 = 0.0007$$

$$Error_{t_p} = (0.0083/37.0219) * 100 = 0.0224$$

CIRCUIT DIAGRAMS & WAVEFORMS

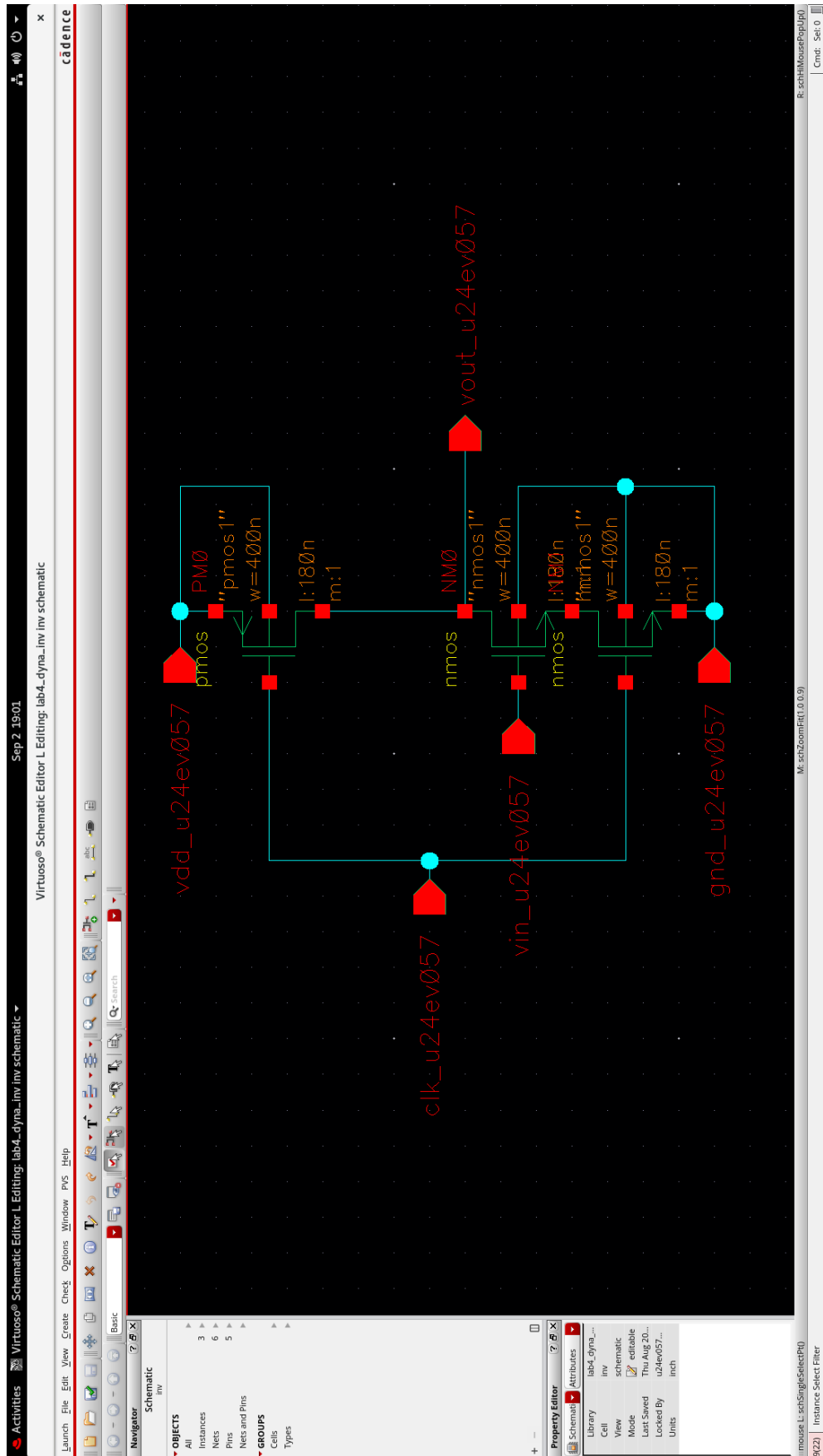


Figure .1: Schematic

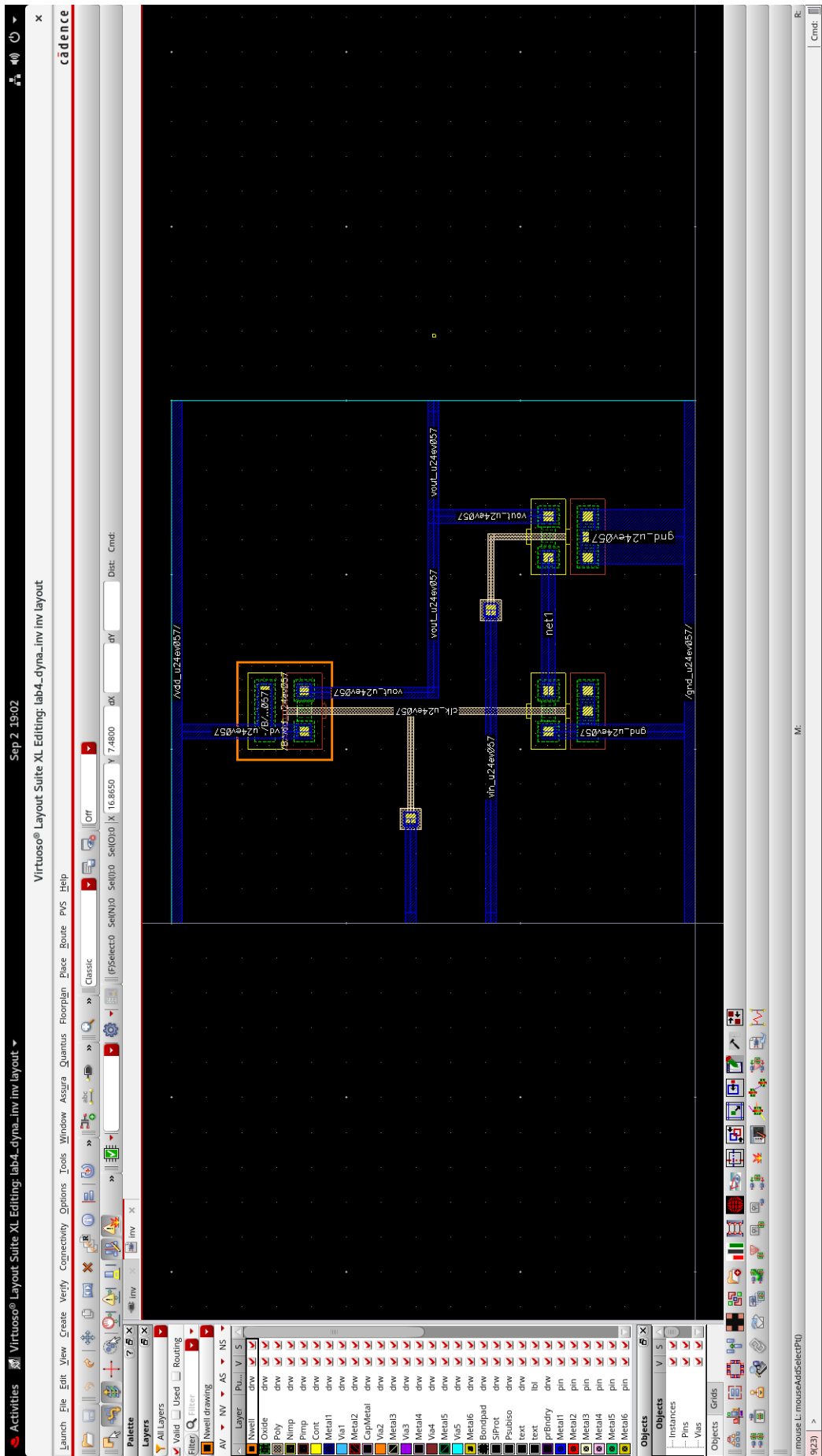


Figure .2: Layout

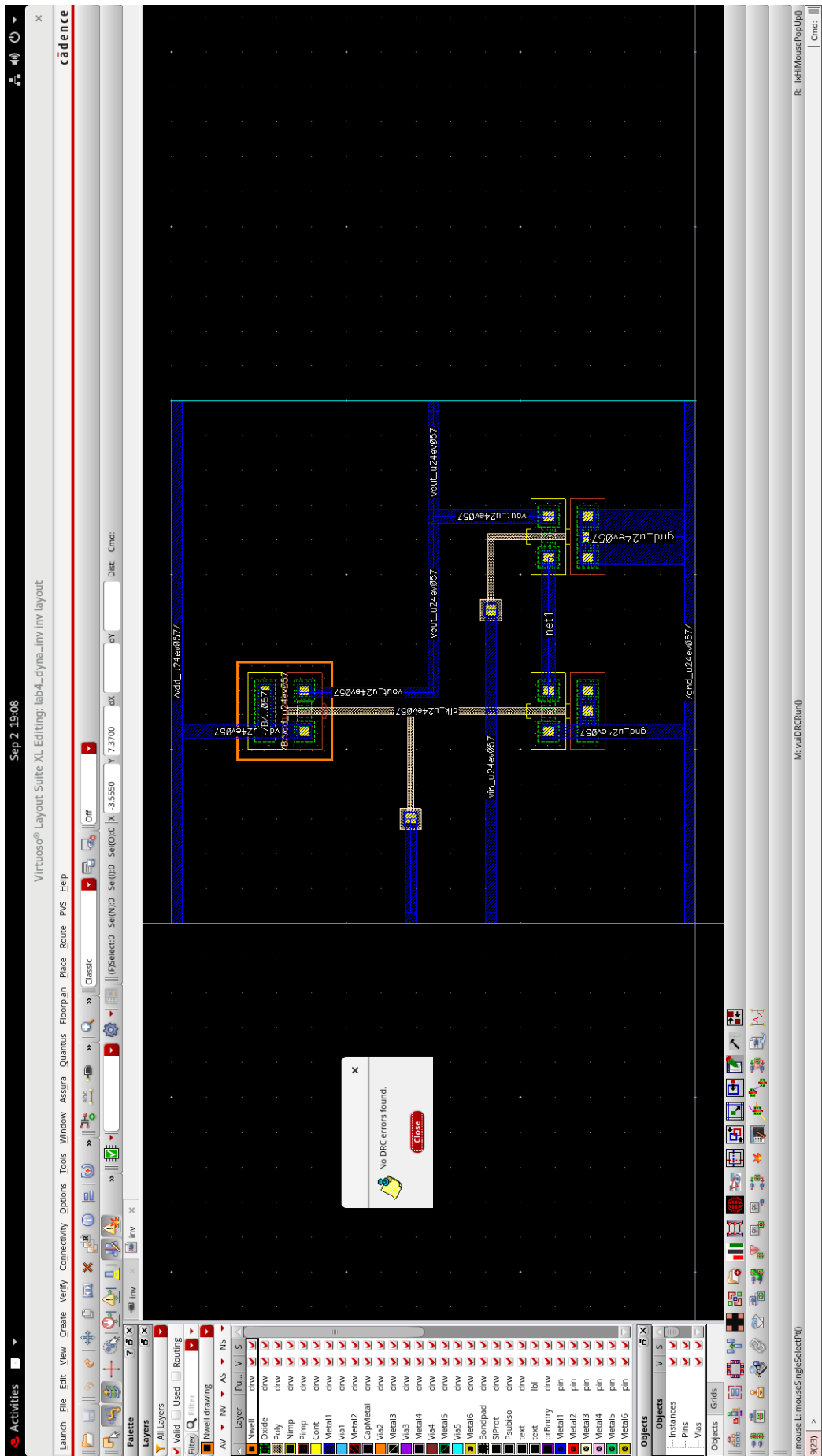


Figure .3: Drc

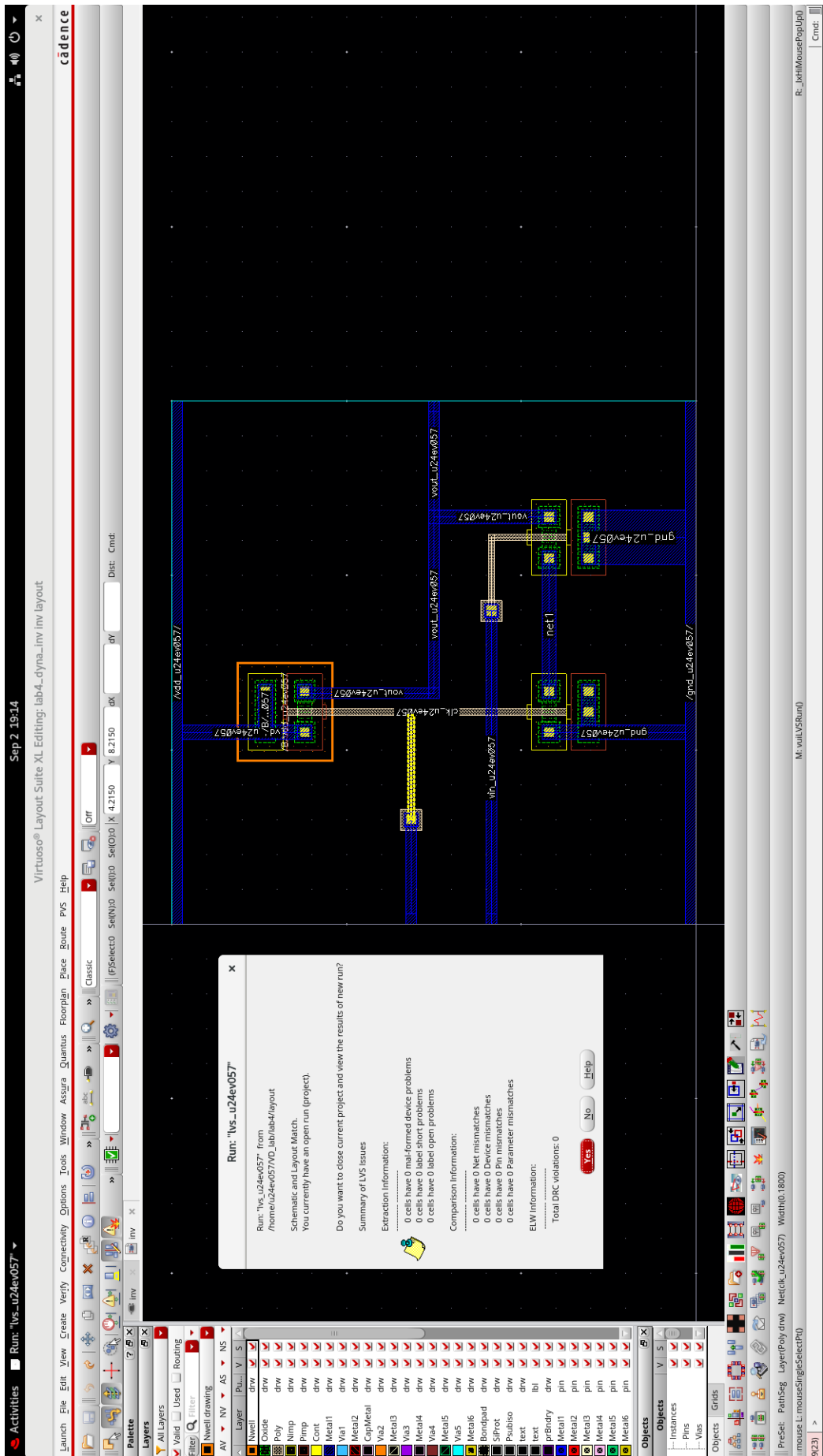


Figure .4: Lvs

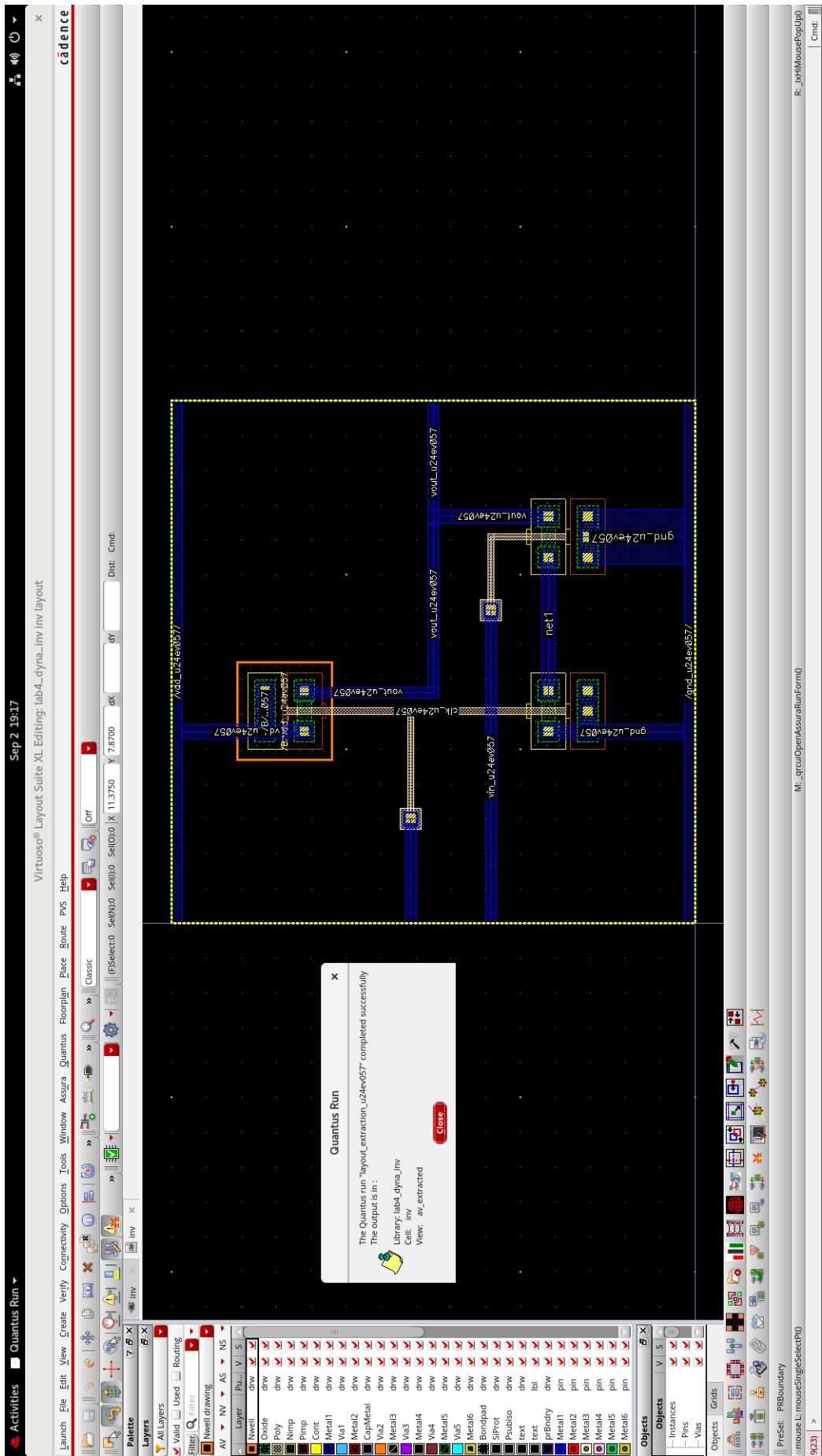


Figure .5: Quantus

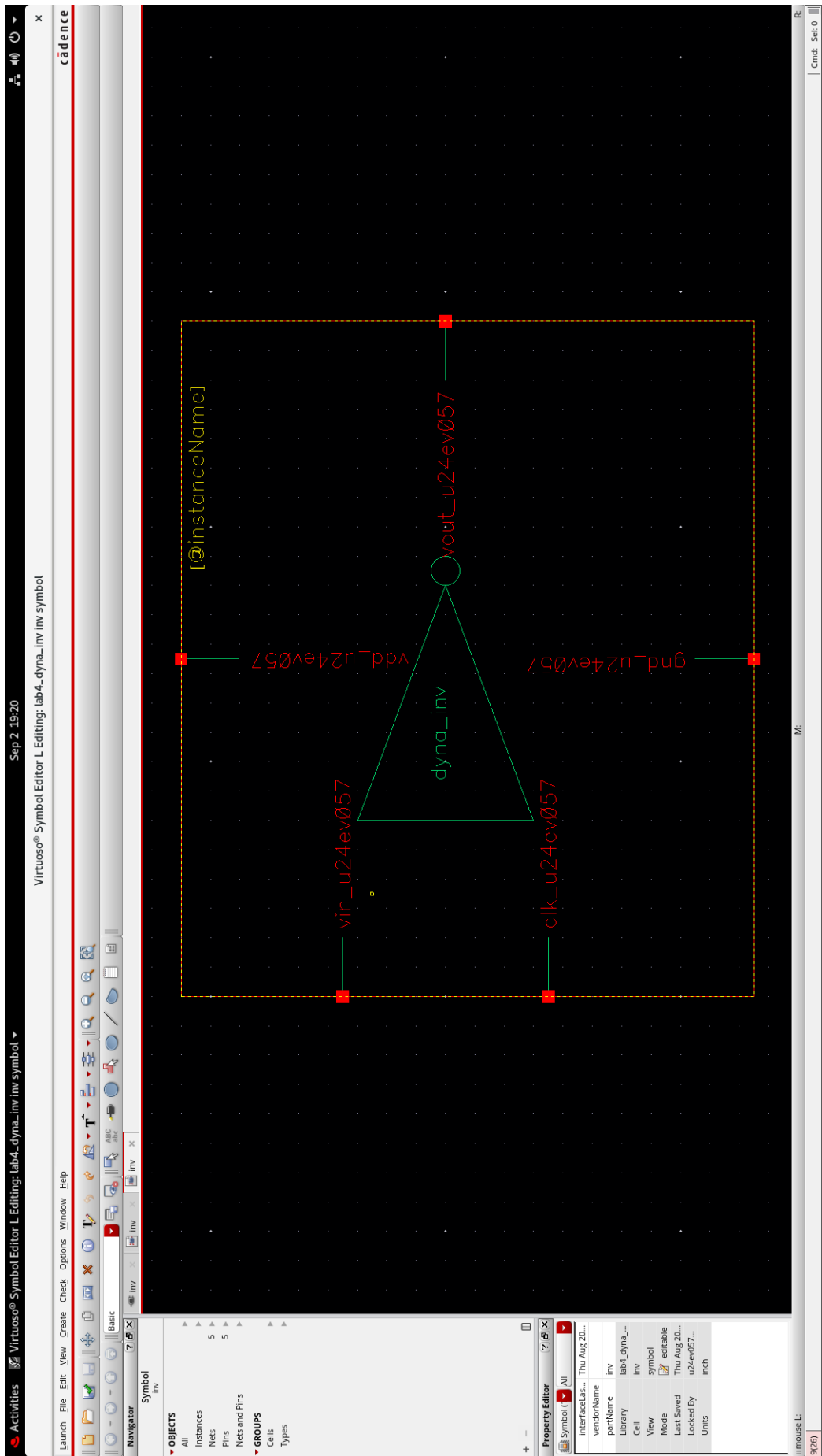


Figure .6: Symbol

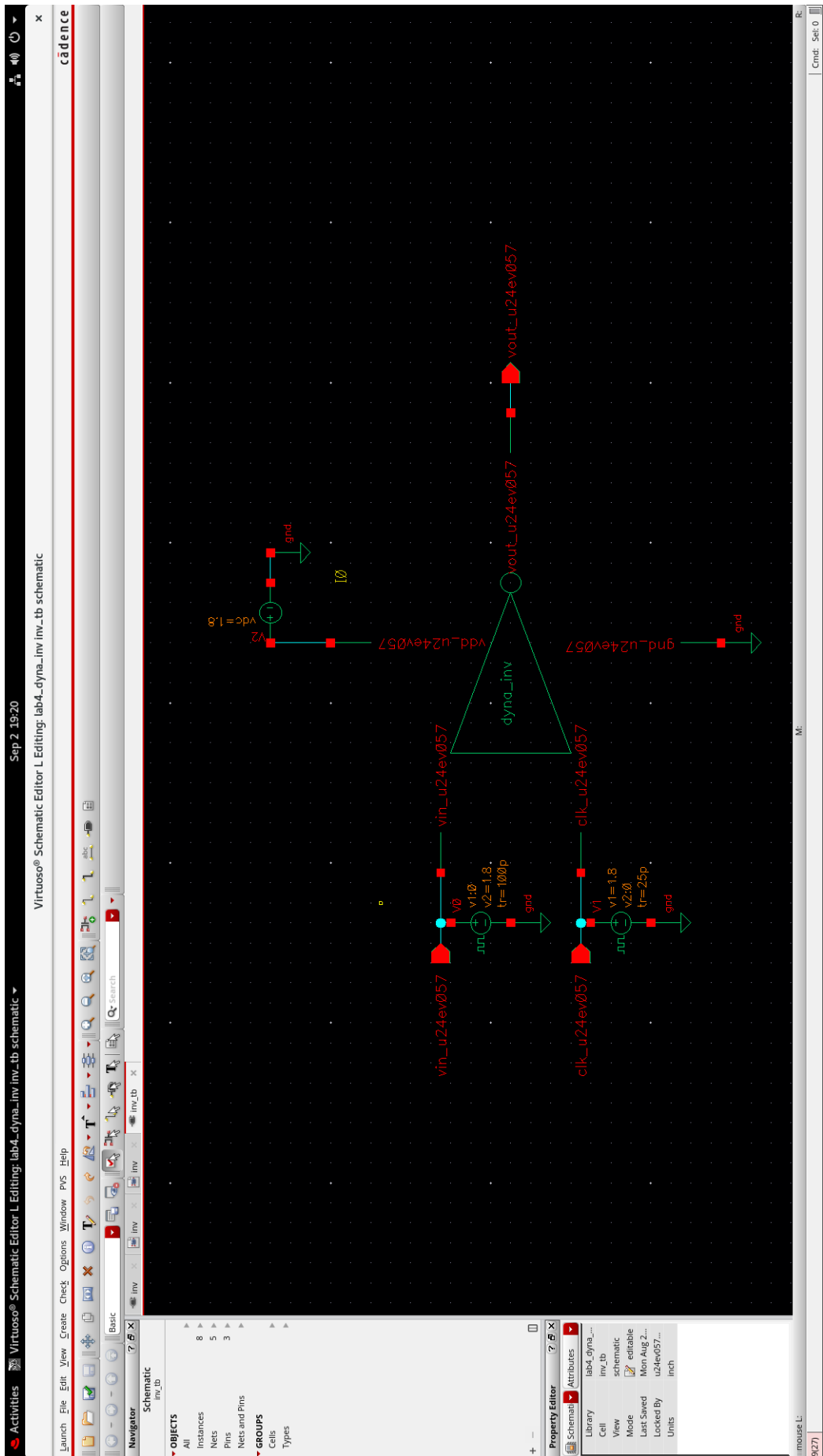


Figure .7: Tb

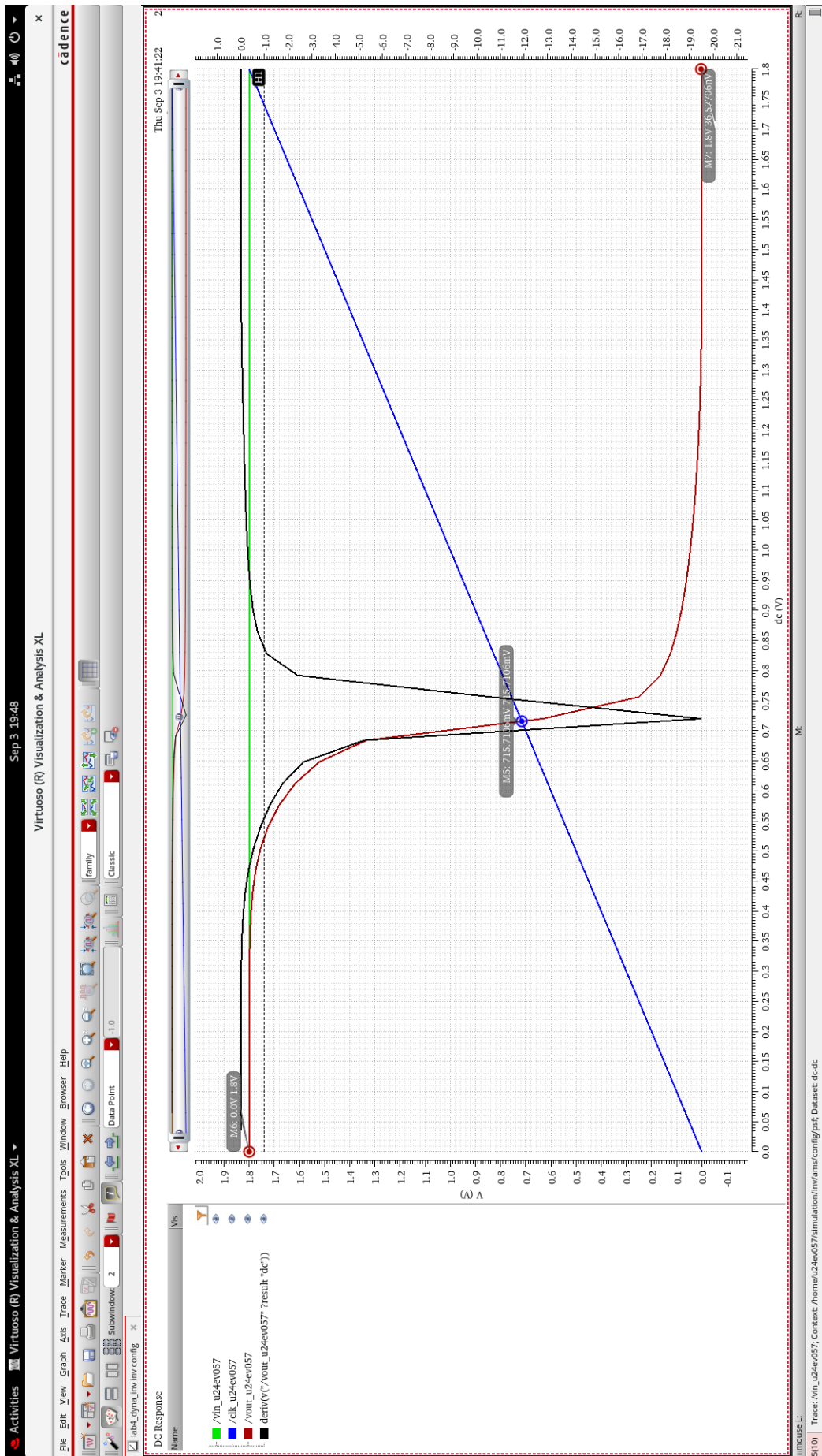


Figure .8: Dc Pre Layout

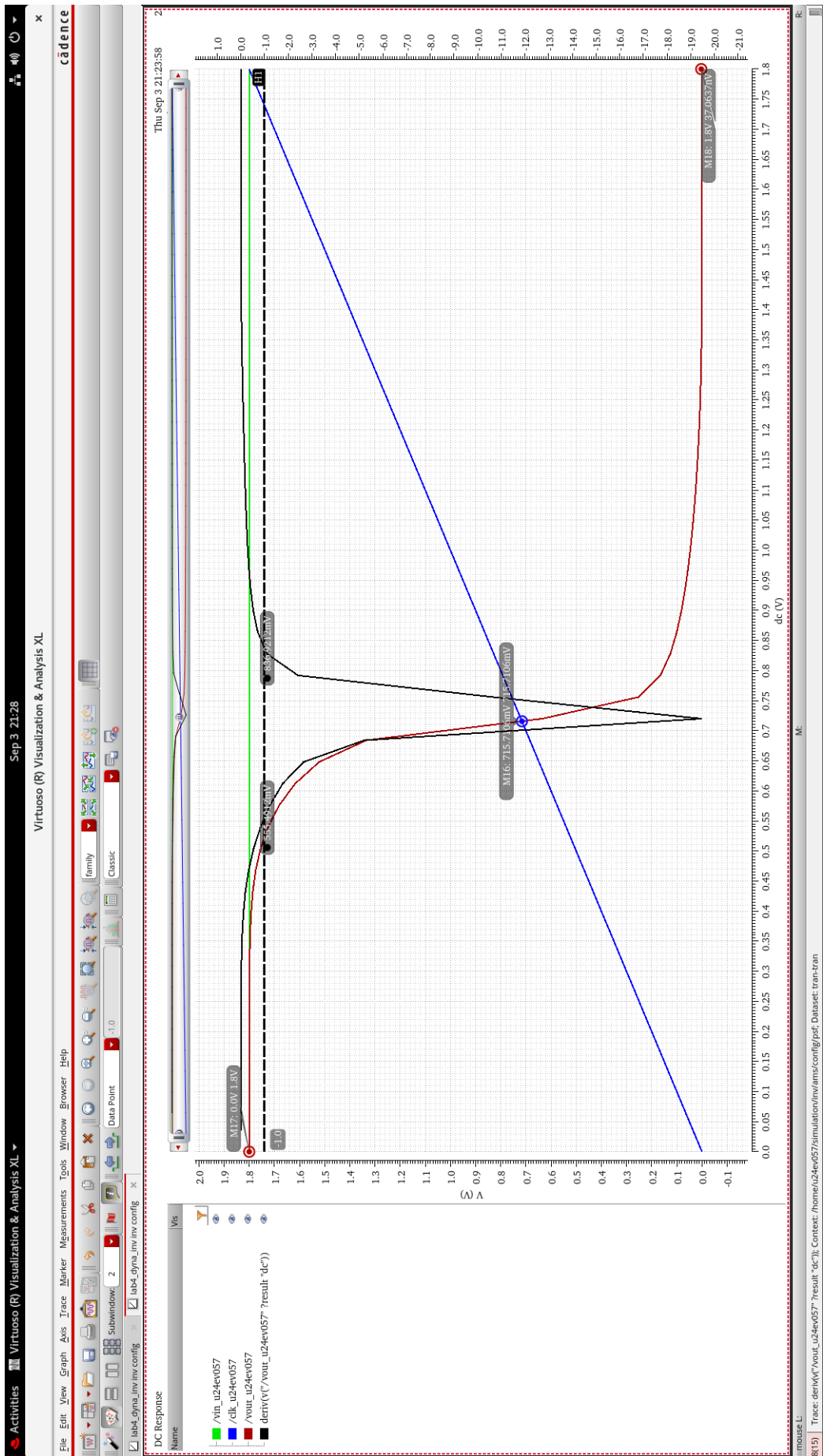


Figure .9: Dc Post Layout

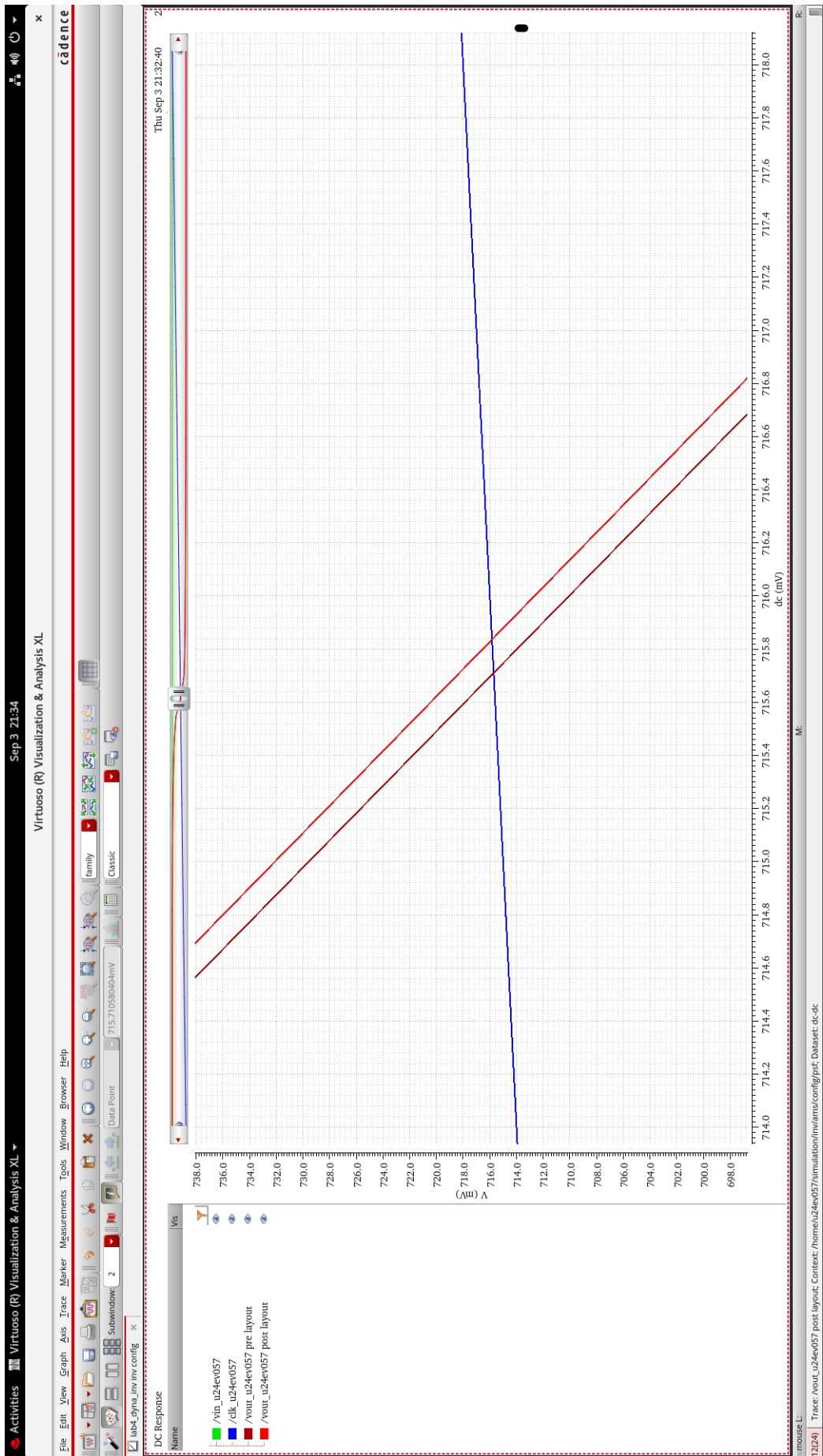


Figure .10: Dc Pre Vs Post

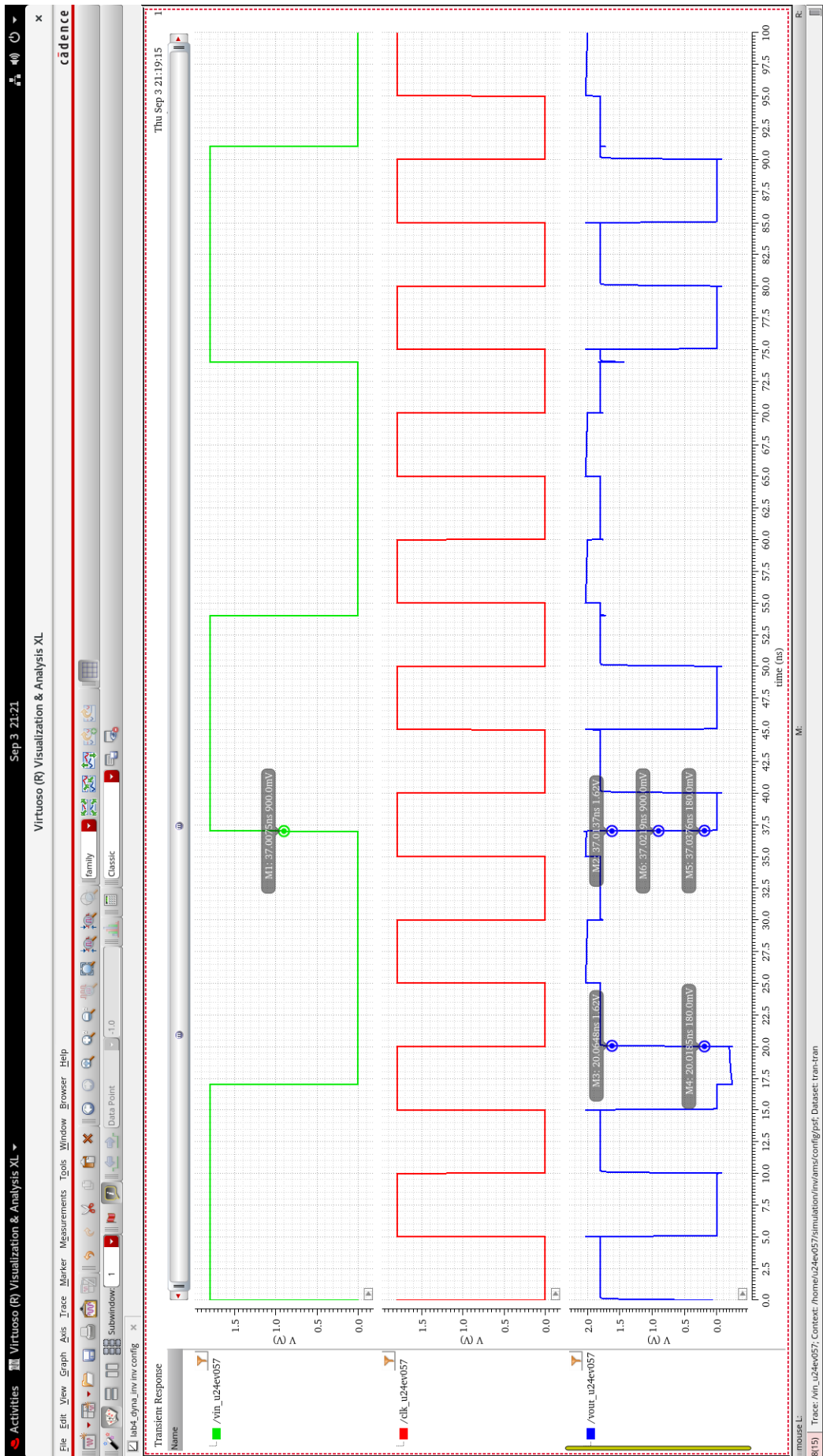


Figure .11: Tran Pre Layout

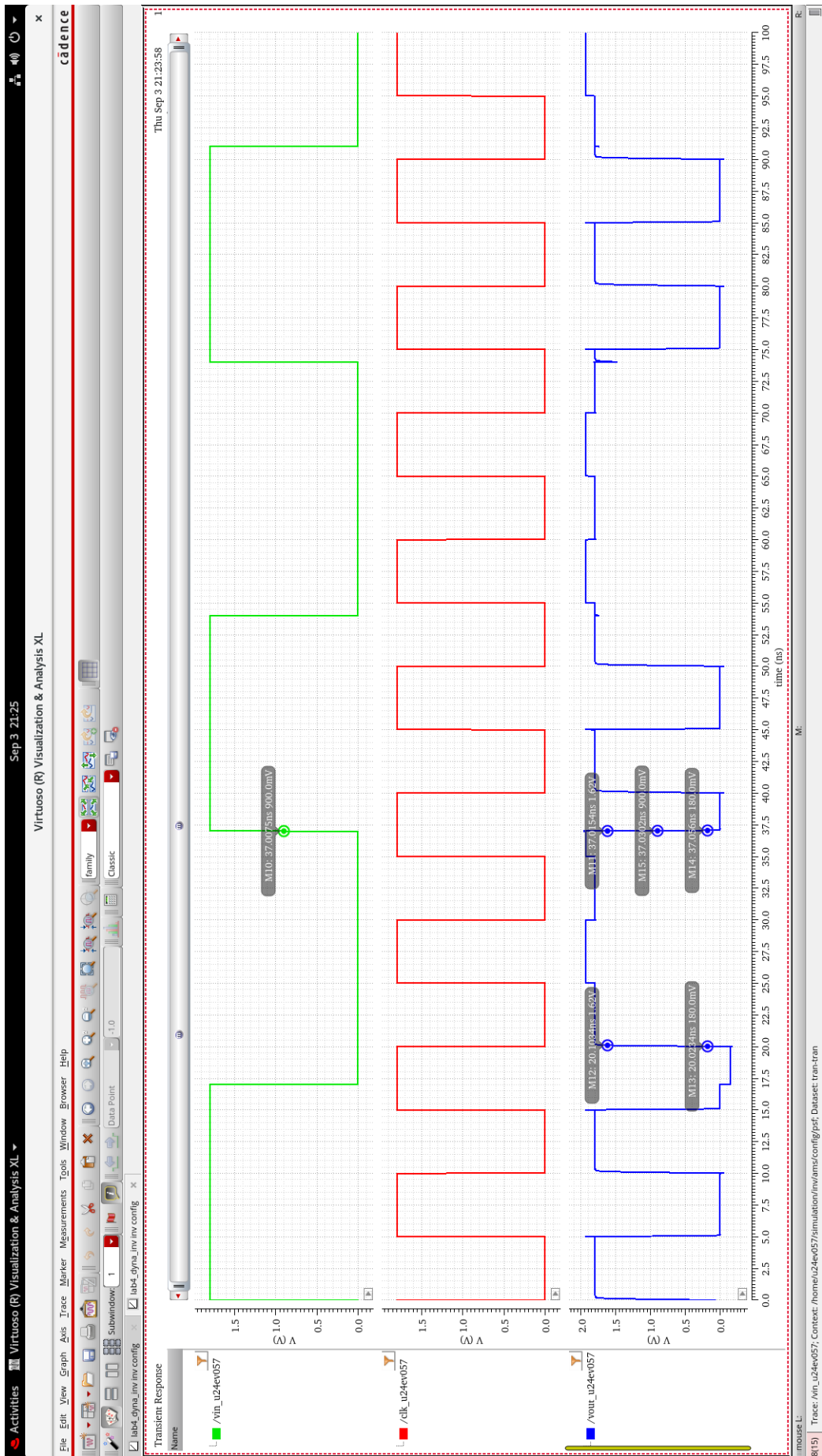


Figure .12: Tran Post Layout

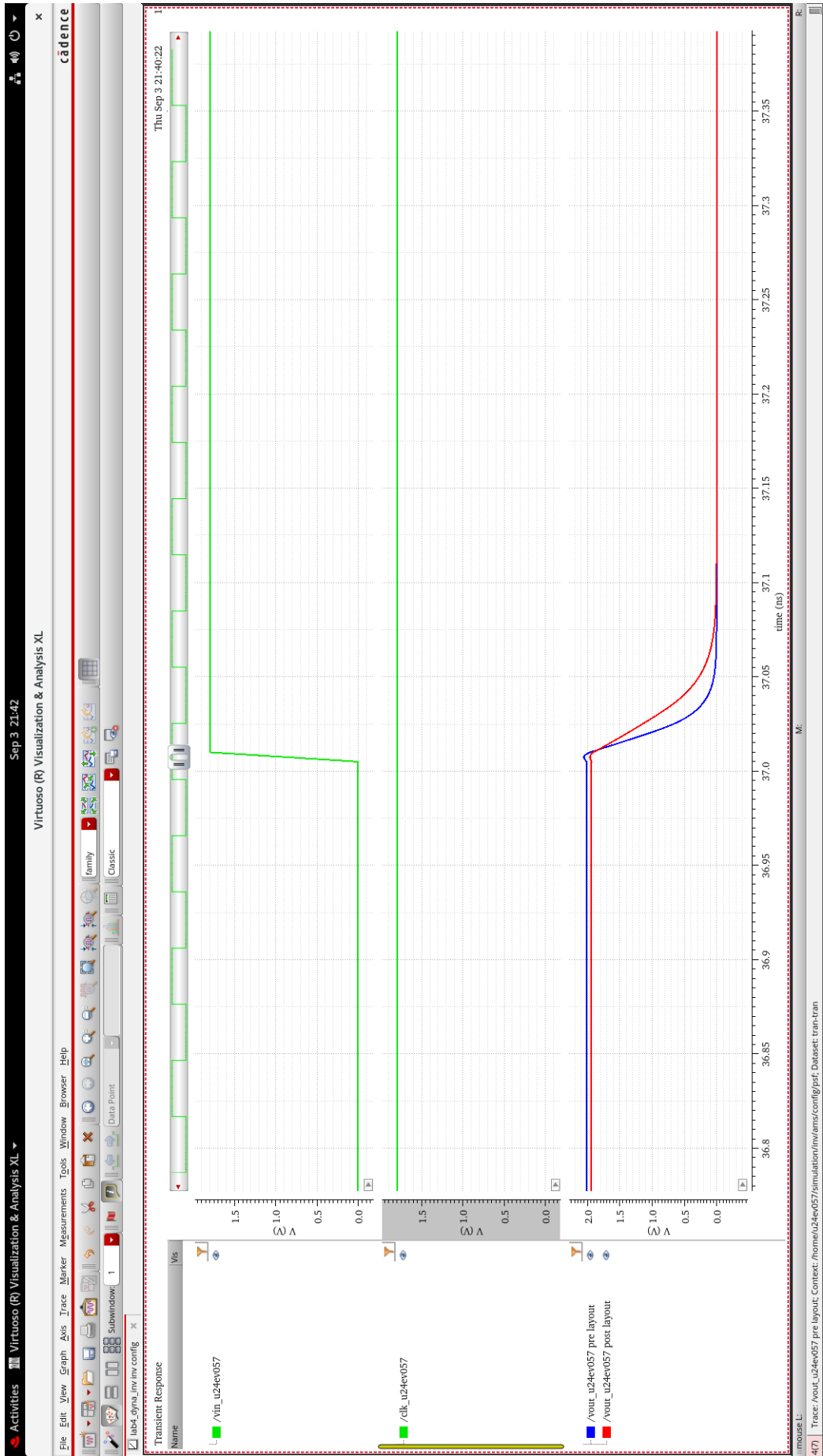


Figure .13: Tran Pre Vs Post

7. RESULT

The static and dynamic CMOS inverter circuits were successfully designed and verified using 180 nm CMOS technology. The simulated parameters and timing characteristics were obtained, and the results were compared to evaluate the performance of both inverter configurations